

Complete Analysis of Compound Mechanism

Engineering and Materials Science

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Sample Mechanism Provided

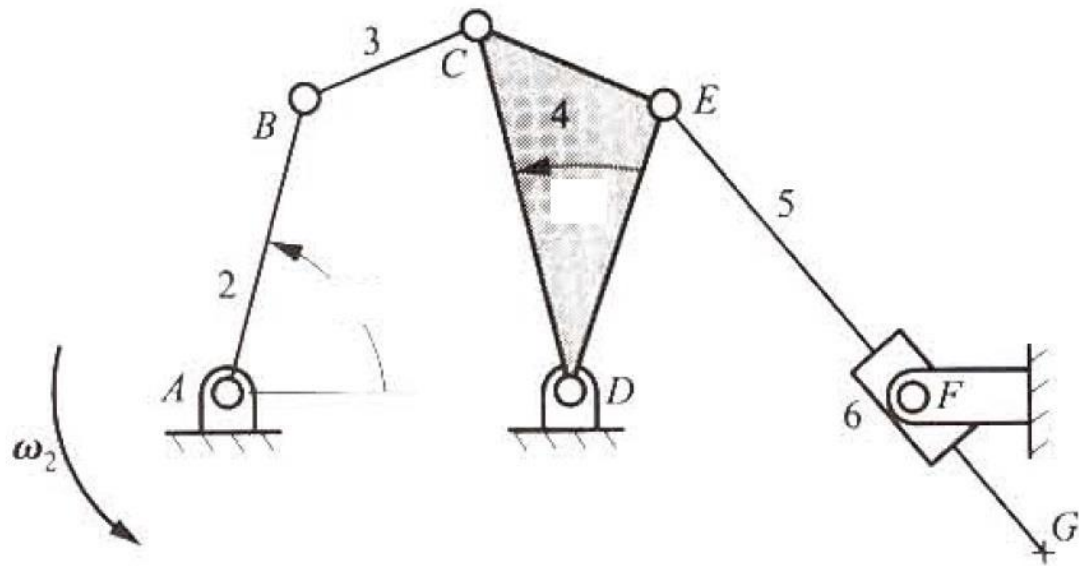


Figure 1: Sample Mechanism

Design of Mechanism on Linkage

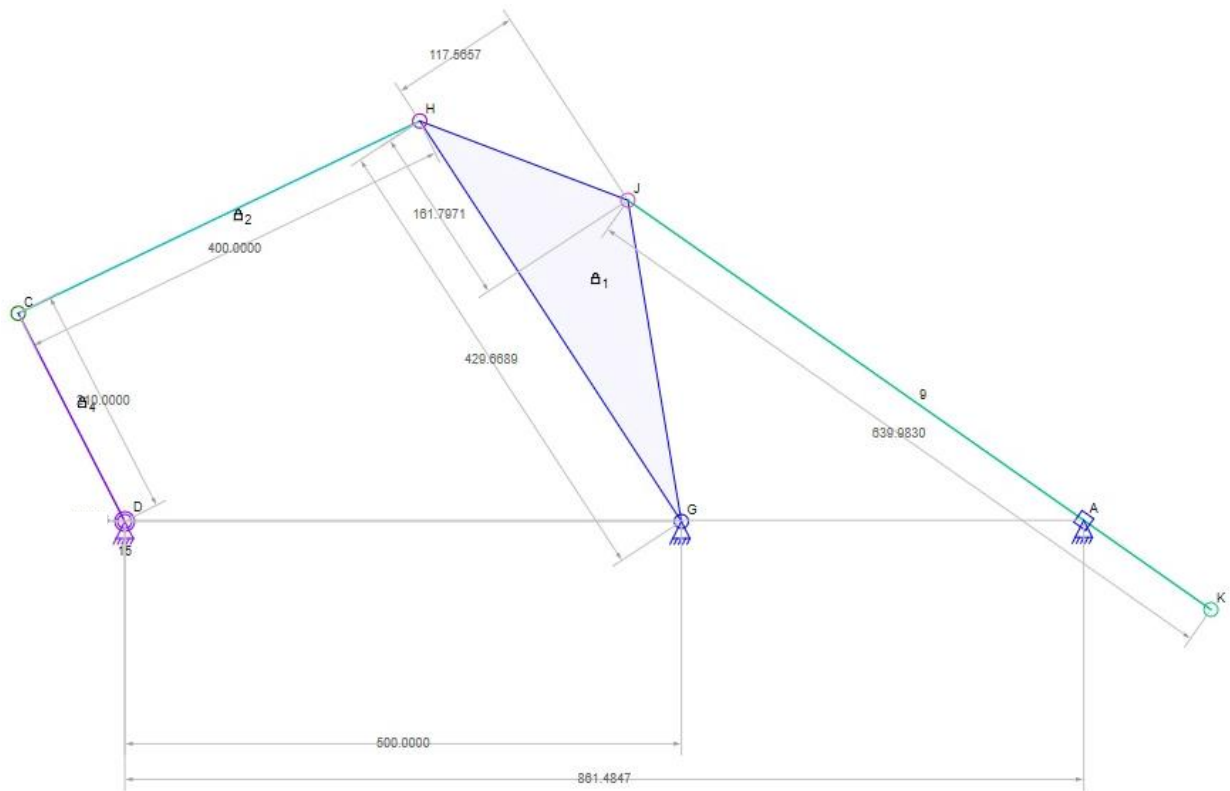


Figure 2: Linkage Mechanism

Analytical Kinematics calculations

Position Analysis

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$\vec{r}_1 + \vec{r}_2 = \vec{r}_3 + \vec{r}_4$

$r_1 e^{j\theta_1} + r_2 e^{j\theta_2} = r_3 e^{j\theta_3} + r_4 e^{j\theta_4}$ $\theta_3 = 0$

$r_1 \cos \theta_1 + j r_1 \sin \theta_1 + r_2 \cos \theta_2 + j r_2 \sin \theta_2 = r_3 \cos \theta_3 + j r_3 \sin \theta_3 + r_4 \cos \theta_4 + j r_4 \sin \theta_4$

real: $r_1 \cos \theta_1 + r_2 \cos \theta_2 = r_3 + r_4 \cos \theta_4$ $r_1 = 210 \text{ mm}$
 imag: $r_1 \sin \theta_1 + r_2 \sin \theta_2 = r_4 \sin \theta_4$ $r_2 = 400 \text{ mm}$
 $r_3 = 500 \text{ mm}$
 $r_4 = 430 \text{ mm}$
 $\theta_3 = 0$
 θ_1

Sol by solving

unknown: θ_1, θ_4

$\vec{r}_5 + \vec{r}_6 = \vec{r}_7$

$r_5 e^{j\theta_5} + r_6 e^{j\theta_6} = r_7 e^{j\theta_7}$

$r_5 \cos \theta_5 + j r_5 \sin \theta_5 + r_6 \cos \theta_6 + j r_6 \sin \theta_6 = r_7 \cos \theta_7 + j r_7 \sin \theta_7$

real: $r_5 \cos \theta_5 + r_6 \cos \theta_6 = r_7$ $r_5 = 292.5$
 imag: $r_5 \sin \theta_5 + r_6 \sin \theta_6 = 0$ $r_7 = 361.5$
 $\theta_7 = 0$
 $\theta_5 = \theta_6 - \phi$

$\phi = \cos^{-1} \left(\frac{r_5^2 - 629.5^2 - 292.5^2}{-2 \times 629.5 \times 292.5} \right)$
 $= 23.7$

unknown: θ_6, θ_5

Figure 3: Position Analysis

Velocity Analysis

Velocity Analysis

$$\underbrace{j\dot{\theta}_1}_{\omega_1} e^{j\theta_1} + \omega_1 r_1 j e^{j\theta_1} + \underbrace{j\dot{\theta}_2}_{\omega_2} e^{j\theta_2} + \omega_2 r_2 j e^{j\theta_2} = \underbrace{j\dot{\theta}_3}_{\omega_3} e^{j\theta_3} + \underbrace{j\dot{\theta}_4}_{\omega_4} e^{j\theta_4} + \omega_4 r_4 j e^{j\theta_4}$$

$$\omega_1 r_1 j e^{j\theta_1} + \omega_2 r_2 j e^{j\theta_2} = \omega_4 r_4 j e^{j\theta_4}$$

Known: r_1, r_2, r_4
 $\theta_1, \theta_2, \theta_4$
 $\omega_1 = 10 \text{ rad/s}$

Real: $-\omega_1 r_1 \sin \theta_1 - \omega_2 r_2 \sin \theta_2 = -\omega_4 r_4 \sin \theta_4$

imag: $\omega_1 r_1 \cos \theta_1 + \omega_2 r_2 \cos \theta_2 = \omega_4 r_4 \cos \theta_4$ unknown: ω_2, ω_4

$$\underbrace{j\dot{\theta}_5}_{\omega_5} e^{j\theta_5} + \omega_5 r_5 j e^{j\theta_5} + \underbrace{j\dot{\theta}_6}_{\omega_6} e^{j\theta_6} + \omega_6 r_6 j e^{j\theta_6} = \underbrace{j\dot{\theta}_7}_{\omega_7} e^{j\theta_7} + \omega_7 r_7 j e^{j\theta_7}$$

Real: $-\omega_5 r_5 \sin \theta_5 + \dot{\theta}_6 \cos \theta_6 - \omega_6 r_6 \sin \theta_6 = 0$ Known: r_5, r_6
 θ_5, θ_6

imag: $\omega_5 r_5 \cos \theta_5 + \dot{\theta}_6 \sin \theta_6 + \omega_6 r_6 \cos \theta_6 = 0$ $\omega_5 = \omega_6$
 unknown: $\dot{\theta}_6, \omega_6$

Figure 4: Velocity Analysis

Acceleration Analysis

ACC analysis

$$\alpha_1 r_1 e^{j\theta_1} + \omega_1^2 r_1 e^{j\theta_1} - \omega_1^2 r_1 e^{j\theta_1} + \alpha_2 r_2 e^{j\theta_2} + \omega_2^2 r_2 e^{j\theta_2} - \omega_2^2 r_2 e^{j\theta_2}$$

$$= \alpha_4 r_4 e^{j\theta_4} + \omega_4^2 r_4 e^{j\theta_4} - \omega_4^2 r_4 e^{j\theta_4}$$

$$\text{Real: } -\omega_1^2 r_1 \cos \theta_1 - \alpha_2 r_2 \sin \theta_2 - \omega_2^2 r_2 \cos \theta_2 = -\alpha_4 r_4 \sin \theta_4 - \omega_4^2 r_4 \cos \theta_4$$

$$\text{Imj: } -\omega_1^2 r_1 \sin \theta_1 + \alpha_2 r_2 \cos \theta_2 - \omega_2^2 r_2 \sin \theta_2 = \alpha_4 r_4 \cos \theta_4 - \omega_4^2 r_4 \sin \theta_4$$

$$\text{Known: } r_1, r_2, r_4 \quad \text{unknown: } \alpha_2, \alpha_4$$

$$\theta_1, \theta_2, \theta_4$$

$$\omega_1, \omega_2, \omega_4$$

$$\alpha_5 r_5 e^{j\theta_5} + \omega_5^2 r_5 e^{j\theta_5} - \omega_5^2 r_5 e^{j\theta_5} + \ddot{\theta}_6 e^{j\theta_6} + 2\dot{\theta}_6 \omega_6 e^{j\theta_6} + \alpha_6 r_6 e^{j\theta_6}$$

$$- \omega_6^2 r_6 e^{j\theta_6} = 0$$

$$\text{real: } -\alpha_5 r_5 \sin \theta_5 - \omega_5^2 r_5 \cos \theta_5 + \ddot{\theta}_6 \cos \theta_6 - 2\dot{\theta}_6 \omega_6 \sin \theta_6 - \alpha_6 r_6 \sin \theta_6 - \omega_6^2 r_6 \cos \theta_6$$

$$\text{mj: } \alpha_5 r_5 \cos \theta_5 - \omega_5^2 r_5 \sin \theta_5 + \ddot{\theta}_6 \sin \theta_6 + 2\dot{\theta}_6 \omega_6 \cos \theta_6 + \alpha_6 r_6 \cos \theta_6 - \omega_6^2 r_6 \sin \theta_6$$

$$\text{Known: } r_5, \theta_5, \theta_6 \quad \text{unknown: } \alpha_6, \ddot{\theta}_6$$

$$\omega_5, \omega_6$$

$$\alpha_5 = \alpha_4$$

Figure 5: Acceleration Analysis

MATLAB computation

```
function w=position(output, input)
```

```
w=[210*cos(input*pi/180) + 400*cos(output(1)) - 500 - 430*cos(output(2));  
    210*sin(input*pi/180) + 400*sin(output(1)) - 430*sin(output(2))];
```

```
end
```

```
function w2=position2(output, input)
```

```
w2=[ 292.5*cos(input*pi/180) + output(1)*cos(output(2)) - 361.5 ;  
     292.5*sin(input*pi/180) + output(1)*sin(output(2)) ];
```

```
end
```

```
function v=velocity(output, input)
```

```
r1 = 210;  
r2 = 400;  
r3 = 500;  
r4 = 430;  
r5 = 292.5;  
r7 = 361.5;  
w1 = 10;
```

```
v=[ -w1*r1*sin(input(1)) - output(1)*r2*sin(input(2)) + output(2)*r4*sin(input(3));  
     w1*r1*cos(input(1)) + output(1)*r2*cos(input(2)) - output(2)*r4*cos(input(3))];
```

```
end
```

```

function v2=velocity2(output, input)

r1 = 210;
r2 = 400;
r3 = 500;
r4 = 430;
r5 = 292.5;
r7 = 361.5;
w1 = 10;

v2=[ -input(1)*r5*sin(input(2)) + output(1)*cos(input(3)) -
output(2)*input(4)*sin(input(3));
      input(1)*r5*cos(input(2)) + output(1)*sin(input(3)) +
output(2)*input(4)*cos(input(3))];
end

```

```

function a=acc1(output, input)

r1 = 210;
r2 = 400;
r4 = 430;
w1 = 10;
w2 = input(1);
w4 = input(2);
theta1 = input(3);
theta2 = input(4);
theta4 = input(5);

a=[ -output(1)*r2*sin(theta2) - w2*w2*r2*cos(theta2) - w1*w1*r1*cos(theta1) +
output(2)*r4*sin(theta4) + w4*w4*r4*cos(theta4);
      output(1)*r2*cos(theta2) - w2*w2*r2*sin(theta2) - w1*w1*r1*sin(theta1) -
output(2)*r4*cos(theta4) - w4*w4*r4*sin(theta4)];

end

```

```

function a2=acc2(output, input)

r5 = 292.5;
r6 = input(1);
r6dot = input(2);
w5 = input(3);
w6 = input(4);
theta5 = input(5);
theta6 = input(6);
a5 = input(7);

a2=[ -a5*r5*sin(theta5) - w5*w5*r5*cos(theta5) + output(2)*cos(theta6) -
2*r6dot*w6*sin(theta6) - output(1)*r6*sin(theta6) - w6*w6*r6*cos(theta6);
      a5*r5*cos(theta5) - w5*w5*r5*sin(theta5) + output(2)*sin(theta6) +
2*r6dot*w6*cos(theta6) + output(1)*r6*cos(theta6) - w6*w6*r6*sin(theta6)];
end

```

```

close
clc % Clears command window

i = 1; % Index initialization
options = optimset('display', 'off');

% Preallocate matrices for efficiency and to fix sizes
output = zeros(1, 361); % 1x361 (for theta values)
theta24 = zeros(2, 361); % 2x361 (Theta2 and Theta4 results)
theta5 = zeros(1, 361); % 1x361 (Theta5 results)
r6theta6 = zeros(2, 361); % 2x361 (r6 and Theta6 results)
w2w4 = zeros(2, 361); % 2x361 (w2 and w4 results)
rdot6w6 = zeros(2, 361); % 2x361 (rdot6 and w6 results)
a2a4 = zeros(2, 361); % 2x361 (a2 and a4 results)
a6r6dotdot = zeros(2, 361); % 2x361 (a6 and r6 double dot results)

for theta = 0:1:360
    output(i) = theta; % Store current theta value

    % Solve position equations
    theta24(:, i) = fsolve(@position, [1 1], options, theta) * 180 / pi; % Store
    % Theta2 and Theta4
    theta5(i) = theta24(2, i) - 23.7; % Compute
    % Theta5

    % Solve position2 equations
    r6theta6(:, i) = fsolve(@position2, [400, 1], options, theta5(i)); % Store r6
    % and Theta6

    r6theta6(2,i) = r6theta6(2,i)*180/pi;

    % solve velocity1 equations
    input1 = [theta*pi/180, theta24(1,i)*pi/180, theta24(2,i)*pi/180];
    w2w4(:,i) = fsolve(@velocity, [1 1], options, input1); % Store w2 and w4

    % solve velocity2 equations
    input2 = [w2w4(2,i), theta5(i)*pi/180, r6theta6(2,i)*pi/180, r6theta6(1,i)];
    rdot6w6(:,i) = fsolve(@velocity2, [1 1], options, input2); % Store r6 dot and
    % w6

    %solve acc1 equations
    input3 = [w2w4(1,i), w2w4(2,i), theta*pi/180, theta24(1,i)*pi/180,
    theta24(2,i)*pi/180];
    a2a4(:,i) = fsolve(@acc1, [1 1], options, input3); % Store a2 and a4

    %solve acc2 equations
    input4 = [r6theta6(1,i), rdot6w6(1,i), w2w4(2,i), rdot6w6(2,i), theta5(i)*pi/180,
    r6theta6(2,i)*pi/180, a2a4(2,i)];
    a6r6dotdot(:,i) = fsolve(@acc2, [1 1], options, input4); % Store a6 and r6
    % double dot

    i = i + 1; % Increment index

```

```

end

% First Figure: position
figure(1);

subplot(4, 1, 1);
plot(output, theta24(1, :));
xlabel('degrees');
ylabel('Theta 2');
title('Graph 1: Theta 2');

subplot(4, 1, 2);
plot(output, theta24(2, :));
xlabel('degrees');
ylabel('Theta 4');
title('Graph 2: Theta4');

subplot(4, 1, 3);
plot(output, r6theta6(2, :));
xlabel('degrees');
ylabel('Theta 6');
title('Graph 4: Theta6');

subplot(4, 1, 4);
plot(output, r6theta6(1, :));
xlabel('degrees');
ylabel('r6/mm');
title('Graph 5: r6');

% Second Figure: Velocity
figure(2);

subplot(4, 1, 1);
plot(output, w2w4(1, :));
xlabel('degrees');
ylabel('w2');
title('Graph 6: w2');

subplot(4, 1, 2);
plot(output, w2w4(2, :));
xlabel('degrees');
ylabel('w4');
title('Graph 7: w4');

subplot(4, 1, 3);
plot(output, rdot6w6(2, :));
xlabel('degrees');
ylabel('w6');
title('Graph 8: w6');

subplot(4, 1, 4);
plot(output, rdot6w6(1, :));
xlabel('degrees');
ylabel('r6 dot/mms^-1');

```

```

title('Graph 9: r6dot');

% third Figure: acceleration
figure(3);

subplot(4, 1, 1);
plot(output, a2a4(1, :));
xlabel('degrees');
ylabel('a2');
title('Graph 10: a2');

subplot(4, 1, 2);
plot(output, a2a4(2, :));
xlabel('degrees');
ylabel('a4');
title('Graph 11: a4');

subplot(4, 1, 3);
plot(output, a6r6dotdot(1, :));
xlabel('degrees');
ylabel('a6');
title('Graph 12: a6');

subplot(4, 1, 4);
plot(output, a6r6dotdot(2, :));
xlabel('degrees');
ylabel('r6 double dot mm/s^-2');
title('Graph 13: r6 double dot');

```

Graphical Representation of unknowns

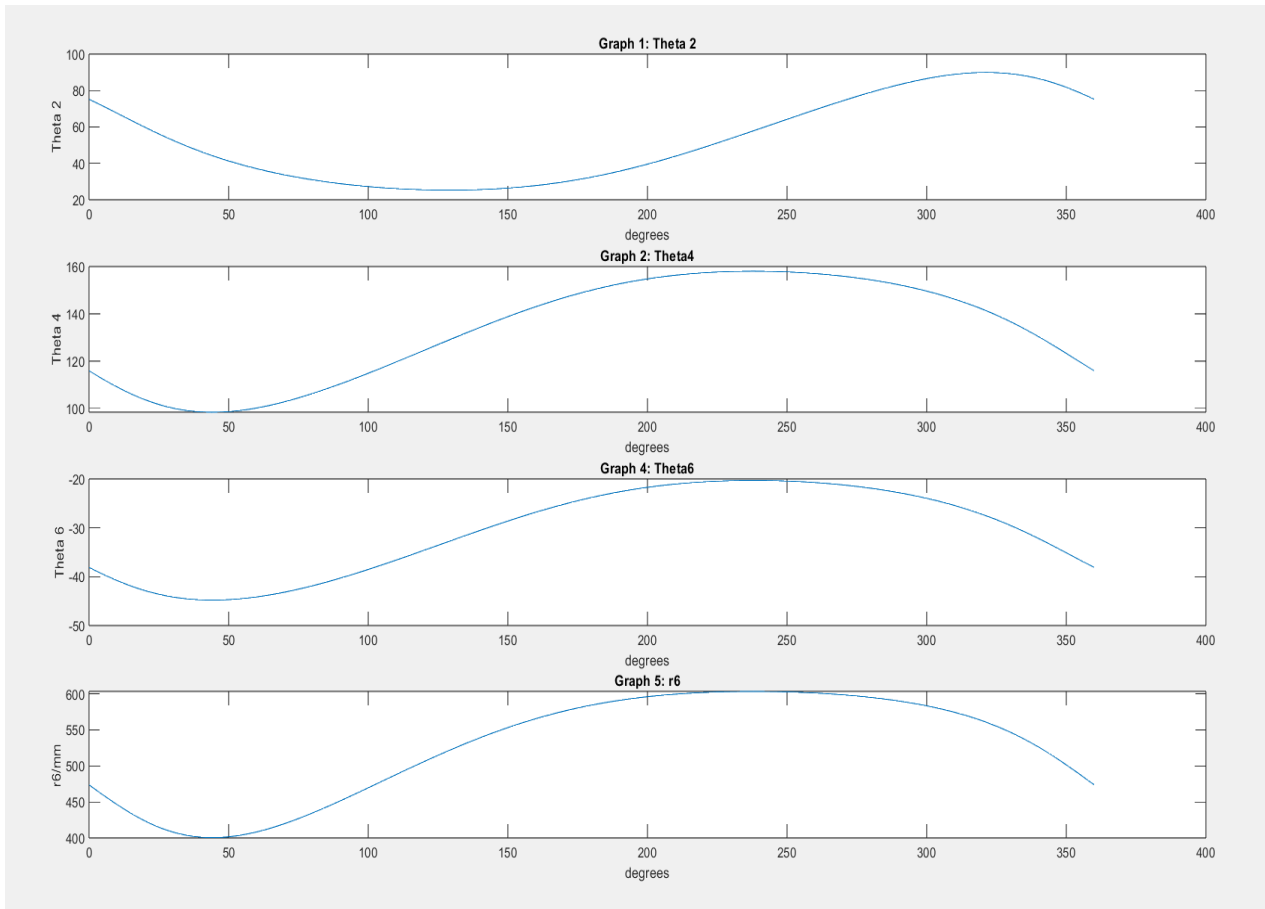


Figure 6: Positional Analysis Graph

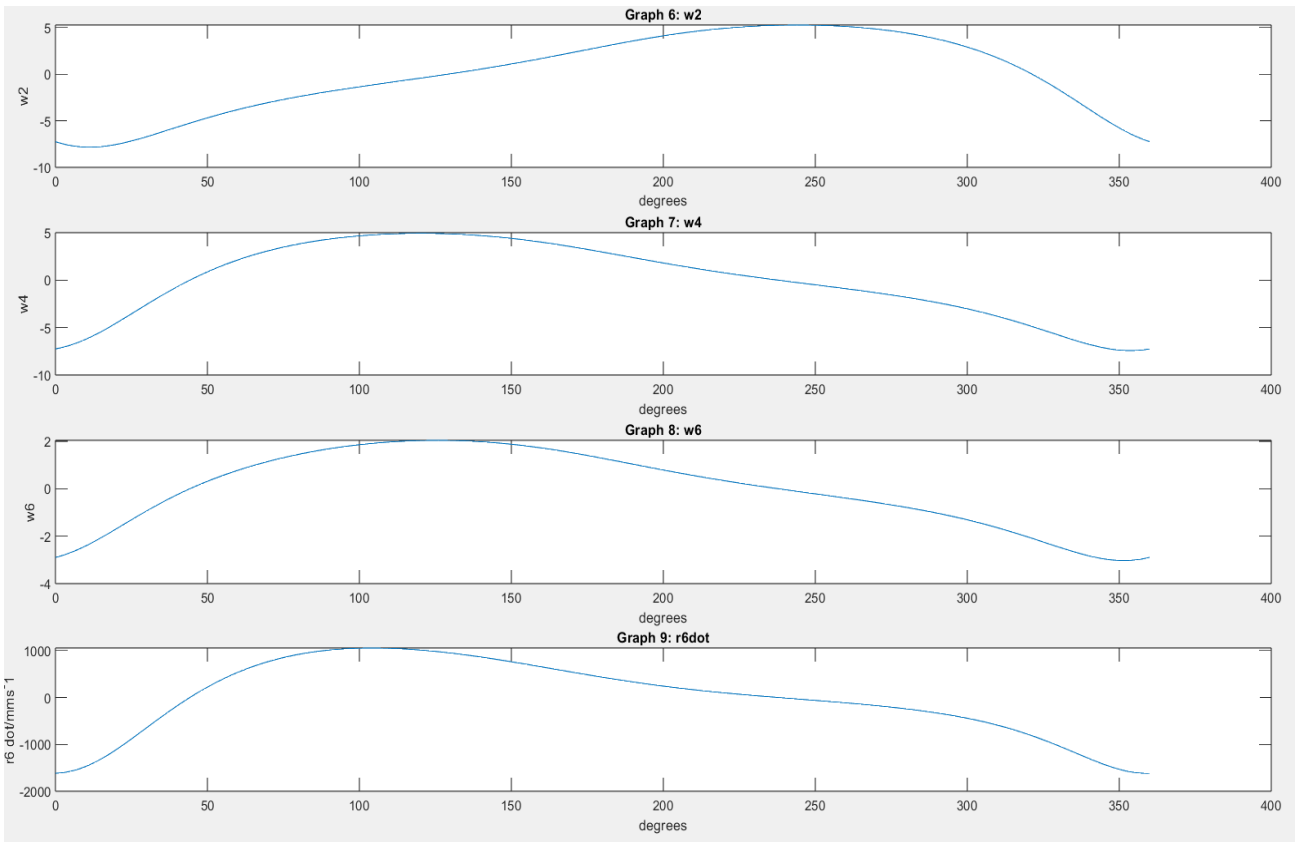


Figure 7: Velocity Analysis Graph

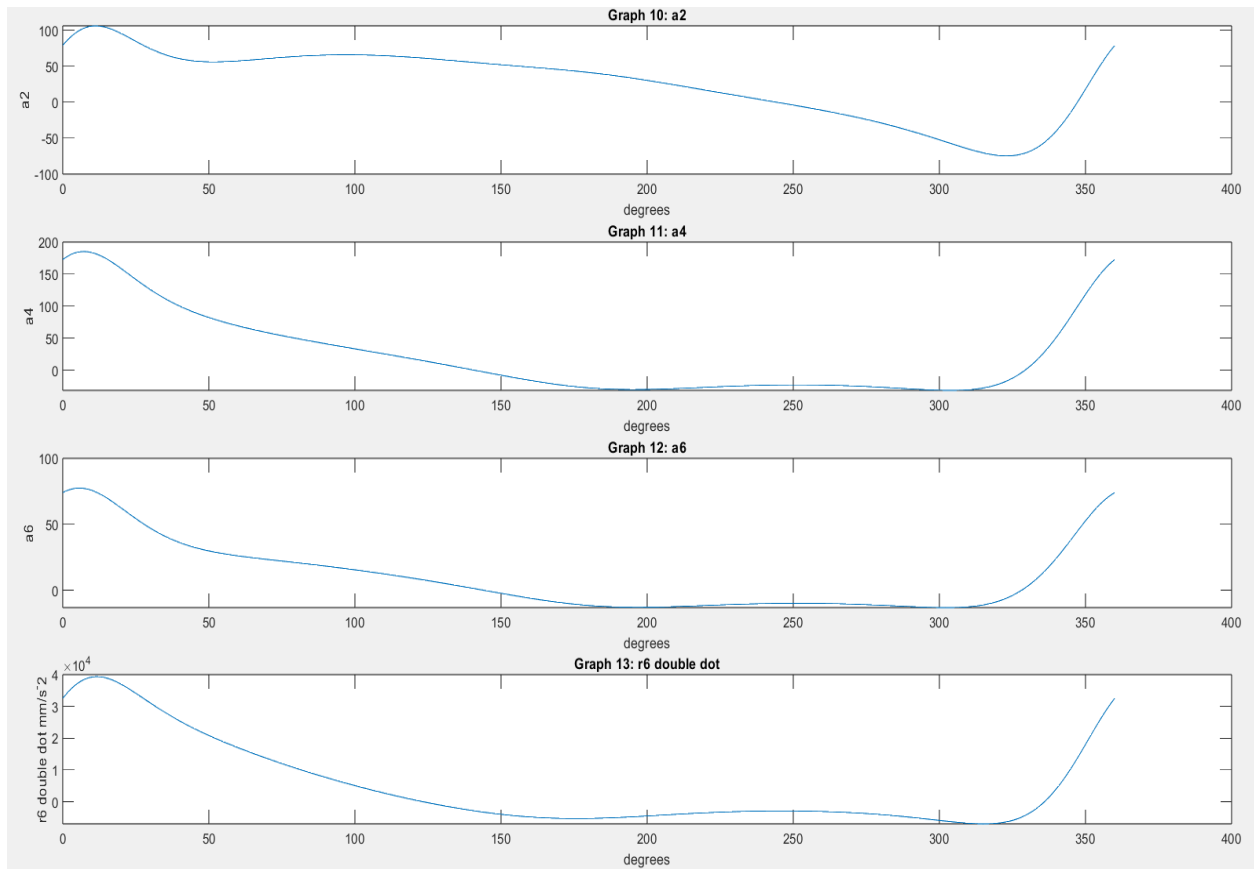


Figure 8: Acceleration Analysis Graph